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The global epidemiology of emerging *Histoplasma* species in recent years

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Abstract: Histoplasmosis is a serious infectious disease in humans caused by *Histoplasma* spp. (Onygenales), whose natural reservoirs are thought to be soil enriched with bird and bat guano. The true global burden of histoplasmosis is underestimated and frequently the pulmonary manifestations are misdiagnosed as tuberculosis. Molecular data on epidemiology of *Histoplasma* are still scarce, even though there is increasing recognition of histoplasmosis in recent years in areas distant from the traditional endemic regions in the Americas. We used multi-locus sequence data from protein coding loci (ADP-ribosylation factor, H antigen precursor, and delta-9 fatty acid desaturase), DNA barcoding (ITS1/2+5.8s), AFLP markers and mating type analysis to determine the genetic diversity, population structure and recognise the existence of different phylogenetic species among 436 isolates of *Histoplasma* obtained globally. Our study describes new phylogenetic species and the molecular characteristics of *Histoplasma* lineages causing outbreaks with a high number of severe outcomes in Northeast Brazil between 2011 and 2015. Genetic diversity levels provide evidence for recombination, common ancestry and clustering of Brazilian isolates at different geographic scales with the emergence of LAm C, a new genotype assigned to a separate population cluster in Northeast Brazil that exhibited low diversity indicative of isolation. The global survey revealed that the high genetic variability among Brazilian isolates along with the presence of divergent cryptic species and/or genotypes may support the hypothesis of Brazil being the center of dispersion of *Histoplasma* in South America, possibly with the contribution of migratory hosts such as birds and bats. Outside Brazil, the predominant species depends on the region. We confirm that histoplasmosis has significantly broadened its area of occurrence, an important feature of emerging pathogens. From a practical point of view, our data point to the emergence of histoplasmosis caused by a plethora of genotypes, and will enable epidemiological analysis focused on understanding the processes that lead to histoplasmosis. Further, the description of this diversity opens avenues for comparative genomic studies, which will allow progress toward a consensus taxonomy, improve understanding of the presence of hybrids in natural populations of medically relevant fungi, test reproductive barriers and to explore the significance of this variation.

Key words: Emerging pathogens, Epidemiology, Genetic diversity, *Histoplasma capsulatum*, Histoplasmosis, Population structure.

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INTRODUCTION

Histoplasmosis is a life-threatening systemic infection caused by the fungal pathogen *Histoplasma capsulatum*. This pathogen was first described in Panama in 1906 by Samuel T. Darling (Darling 1906), followed shortly thereafter by reported infections throughout the Americas (Riley & Watson 1926, Negroni 1940, Furcow 1958) and later in widely scattered places around the world (Rocher 1950, Ashbee *et al.* 2008, Antinori 2014, Baker *et al.* 2019). The first epidemiological investigations in the mid-1940s revealed that Darling's disease was more frequent than previously thought (Christie & Peterson 1945). There is an increasing awareness of histoplasmosis in Central and South America in patients with HIV/AIDS, and deaths from histoplasmosis in this region may outnumber deaths from tuberculosis (Pasqualotto & Queiroz-Telles 2018, Linder & Kauffman 2019, Nacher *et al.* 2019). Currently, histoplasmosis is one of the world's leading systemic infections, with incidence ranging from 0.1 to 1 case per 100 000 inhabitants per year in temperate climates, 10 to 100 cases per 100 000 in the humid tropics, and over 100 cases per 100 000 in high risk groups and during outbreaks (Colombo *et al.* 2011, Adenis *et al.* 2018). Although most infections are not outbreak-associated, outbreaks of histoplasmosis linked to a common source do occasionally occur and often involve activities that disrupt soil, especially samples that contains bird or bat droppings (Queiroz-Telles *et al.* 2017).

Mammals become infected after inhaling *Histoplasma* propagules from the environment (Deepe 2018), which leads, in the vast majority of human cases, to a primary pulmonary infection that is unapparent, subclinical or completely benign (Eissenberg & Goldman 1991). The remaining patients may develop chronic progressive lung disease, chronic cutaneous or systemic disease, or an acute fulminating, rapidly fatal, systemic infection (Kauffman 2007, Oladele *et al.* 2018). The infection is also very common in wild (e.g., marsupials, rodents, armadillos, sloths, bats) and domestic animals (e.g., dogs and cats) in endemic areas (Emmons 1950, Seyedmousavi *et al.* 2018). However,

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mammals appear to be dead-end hosts of *Histoplasma* since there is no person-to-person or animal-to-person spread.

For over a century histoplasmosis was attributed to a single etiological agent, *H. capsulatum* (Onygenales). Judging from different clinical manifestations, morphologies and wide-ranging geographical occurrence, three different varieties are historically recognised: (i) *H. capsulatum* var. *capsulatum*, which is a broadly dispersed New World human pathogen, responsible for classic histoplasmosis (Eissenberg & Goldman 1991); (ii) *H. capsulatum* var. *duboisii*, a human pathogen confined to Central and West Africa, causing African histoplasmosis, whose principal features are skin or bone lesions (Loulergue *et al.* 2007, Pakasa *et al.* 2018); and (iii) *H. capsulatum* var. *farciminosum*, an Old World animal pathogen, responsible for epizootic lymphangitis in equines (Selim *et al.* 1985).

Previous studies exploring phylogeography, gene flow and hybridisation processes have shown that the causal agents of histoplasmosis encompass several cryptic species, which are structured according to their geographical origin (Van Dyke *et al.* 2019). Kasuga and colleagues studied 137 isolates recovered from 25 countries on six continents and were able to recognise at least eight clades within seven phylogenetic species, including: (i) North American class 1 clade (NAm 1); (ii) North American class 2 clade (NAm 2); (iii) Latin American group A clade (LAm A); (iv) Latin American group B clade (LAm B); (v) Australian clade; (vi) Netherlands (Indonesian?) clade; (vii) Eurasian clade and (viii) African clade (Kasuga *et al.* 1999, 2003). Recently, based on the genealogic concordance for phylogenetic species recognition (GCPSR) and population structures methods, six new phylogenetic species were proposed: LAm A1; LAm A2; RJ (which corresponds to the LAm A clade described by Kasuga *et al.*, 2003); LAm B1; LAm B2 (corresponding to LAm B clade, (Kasuga *et al.* 2003); and BAC1 (a bat-associated species-specific clade) (Teixeira *et al.* 2016).

A more robust and comprehensive taxonomic scenario was recently proposed for *Histoplasma* based on whole-genome data from 30 isolates embedded in five of the groups proposed by Kasuga *et al.* (Kasuga *et al.* 2003), namely the NAm 1, NAm 2, LAm A, Panama/H81 lineage, and Africa groups. Sepúlveda and colleagues demonstrated the presence of independently evolving lineages, which were elevated to species status and named *H. capsulatum sensu stricto* (*s. str.*) when referring to the Panamanian lineage (H81 lineage), *Histoplasma mississippiense* sp. nov. (formerly known as NAm 1), *Histoplasma ohiense* sp. nov. (formerly known as NAm 2) and *Histoplasma suramericanum* sp. nov. (formerly known as LAm A) (Sepúlveda *et al.* 2017).

Although *Histoplasma* occurs globally, it appears that the phylogenetic species are not evenly distributed. For example, infections due to *H. mississippiense* and *H. ohiense* are more common in the USA, where they occur in syntopy. Also, *H. capsulatum s. str.* and *H. suramericanum* coexist in sympatry, since their geographical distributions overlap in central and possibly northwestern Colombia (Sepúlveda *et al.* 2017, Gomez *et al.* 2019, Sahaza *et al.* 2019). Moreover, *H. capsulatum* var. *duboisii*, the agent of African histoplasmosis is found predominantly on continental Africa and Madagascar and is argued to be a separate species (Sepúlveda *et al.* 2017, Valero *et al.* 2018, Van Dyke *et al.* 2019). Along with geographical differences, there is differential virulence among phylogenetically related species of *Histoplasma* (Sepúlveda *et al.* 2014). Whether these differences are responsible for differences in clinical presentations needs to be confirmed in a large group of patients. Previous studies indicate that *H. suramericanum* causes an acute pulmonary disease with a pronounced lung pathology that leads to high mortality, whereas *H. mississippiense* and *H. ohiense* cause a chronic pulmonary disease (Durkin *et al.* 2004, Sepúlveda *et al.* 2017). *H. capsulatum* var. *duboisii* was recently associated with AIDS and disseminated disease and to a lesser extent with skin lesions in immunocompetent persons (Oladele *et al.* 2018, Valero *et al.* 2018).

Several studies at the molecular level have been conducted to elucidate the genetic diversity and population structure of *Histoplasma* using molecular markers, ranging from DNA sequencing to DNA fingerprinting (Damasceno *et al.* 2016, Sahaza *et al.* 2019). In Brazil, high genetic diversity correlates with the geographical origin (Zancope-Oliveira *et al.* 2005, Muniz Mde *et al.* 2010, Almeida *et al.* 2019). However, the origin of the Brazilian *Histoplasma* populations and their relationship with the global populations remain unknown, especially considering the recently proposed taxonomic changes. Hence, the aims of the present study were to understand the genetic diversity of *Histoplasma* isolates in Brazil and explore their evolutionary relationships with *H. capsulatum s. str.*, *H. capsulatum* var. *duboisii*, *H. suramericanum*, *H. mississippiense* and *H. ohiense*. We used multi-locus sequence data from protein coding loci (ADP-ribosylation factor, H antigen precursor, and delta-9 fatty acid desaturase), DNA barcoding (ITS1/2+5.8s), AFLP markers and mating type analysis to determine the genetic diversity, population structure and recognise the existence of different phylogenetic species of *Histoplasma* isolates obtained from Northeast and Southeast Brazil. Our data included a wider global panel of isolates leading us to hypothesise that the high genetic variability among Brazilian isolates along with the presence of divergent cryptic species and/or genotypes support the idea that Brazil is the center of dispersion of *Histoplasma* in South America, possibly with the contribution of migratory hosts such as birds and bats.

MATERIAL AND METHODS

Fungal strains

Isolates were of clinical (n = 102) or veterinary origin (n = 2), from different geographic regions of Brazil, comprising the main endemic areas of histoplasmosis (Table S1). Clinical isolates of *H. capsulatum* were recovered from patients with acute pulmonary, chronic pulmonary and disseminated forms of histoplasmosis. Isolates were stored as slant cultures on Sabouraud dextrose agar (Difco Laboratories, Detroit, MI, USA) at room temperature. The total dataset comprised 474 operational taxonomic units (OTUs) (Table S1).

DNA extraction and molecular characterisation

Total DNA was obtained and purified directly from 14 day-old monosporic colonies on Sabouraud slants by following the Fast DNA kit protocol (MP Biomedicals, USA), as previously described (Rodrigues *et al.* 2014). All isolates were characterised at the species level by PCR using a *Histoplasma*-specific primer pair (Table 1) targeting the gene encoding the M-antigen, by using primers Msp2F and Msp2R (Guedes *et al.* 2003) or by

Table 1. Primers used in this study for generic amplification, sequencing and genotyping.

Locus/Region	Primer	Primer sequence 5′to 3′	Annealing temperature (°C)	Orientation	Amplicon (bp)	Reference
ARF	ARF1	AGAATATGGGGCAAAAAGGA	65–56*	Forward	470	(Kasuga *et al.* 1999)
	ARF2	CGCAATTCATCTTCGTTGAG	65–56*	Reverse		(Kasuga *et al.* 1999)
H-anti	H-anti3	CGCAGTCACCTCCATACTATC	65–56*	Forward	410	(Kasuga *et al.* 1999)
	H-anti4	GCGCCGACATTAACCC	65–56*	Reverse		(Kasuga *et al.* 1999)
OLE	Ole3	TTTAAACGAAGCCCCCACGG	65–56*	Forward	424	(Kasuga *et al.* 1999)
	Ole4	CACCACCTCCAACAGCAGCA	65–56*	Reverse		(Kasuga *et al.* 1999)
ITS	ITS5	GGAAGTAAAAGTCGTAACAAGG	52	Forward	630	(White *et al.* 1990)
	ITS4	TCCTCCGCTTATTGATATGC	52	Reverse		(White *et al.* 1990)
MAT 1-1	MAT1-1S	CGTGGTTAGTTACGGAGGCA	60	Forward	440	(Bubnick & Smulian 2007)
	MAT1-1AS	TGAGGATGCGAGTGATGGGA	60	Reverse		(Bubnick & Smulian 2007)
MAT 1-2	MAT1-2S	ACACAGTAGCCCAACCTCTC	60	Forward	528	(Bubnick & Smulian 2007)
	MAT1-2AS	TCGACAATCCCATCCAATACCG	60	Reverse		(Bubnick & Smulian 2007)
M-anti	Msp2F	CGGGCCGCGTTTAACAGCGCC	55	Forward	279	(Guedes *et al.* 2003)
	Msp2R	ACCAGCGGCCATAAGGACGTC	55	Reverse		(Guedes *et al.* 2003)
SCAR marker	1281-1283$_{220}$F	CATTGTTGGAGGAACCTGCT	55	Forward	220	(Frías De León *et al.* 2012)
	1281-1283$_{220}$R	GAGCTGCAGGATGTTTGTTG	55	Reverse		(Frías De León *et al.* 2012)
SCAR marker	1281-1283$_{230}$F	GGAGCCATGACGTTAAATGG	55	Forward	230	(Frías De León *et al.* 2012)
	1281-1283$_{230}$R	TATTGCCAATGGGTTTGTCA	55	Reverse		(Frías De León *et al.* 2012)

arf: ADP-ribosylation factor; H-anti: H antigen precursor; ole: Delta-9 fatty acid desaturase; ITS: Internal transcribed spacers + rRNA genes; MAT: Mating types; M-anti: M antigen precursor; Scar marker: Sequence-Characterised Amplified Region. *Touchdown PCR

using sequence-characterised amplified region (SCAR) markers 1281-1283$_{220}$ (PCR220) and 1281-1283$_{230}$ (PCR230), as described before (Frías De León *et al.* 2012). Reference strains representing the main phylogenetic groups in *Histoplasma* were included in all experiments (Table S1). Diagnostic values using each primer pair included sensitivity, specificity, positive predictive value (PPV), and negative predictive value (NPV). The receiver operating characteristic (ROC) curves were prepared and analyzed to determine the sensitivity and specificity of each PCR assay (Msp2F-Msp2R, PCR220, PCR230). The area under the ROC curve (AUC) was calculated to evaluate the diagnostic value of each PCR assay. *P*-values $\leq$0.05 were considered statistically significant. To measure the degree of concordance of each PCR assay, we calculated the *Kappa* statistic and its 95 % confidence interval (CI). *Kappa* values were interpreted as follows: 0.00–0.20, poor agreement; 0.21–0.40, fair agreement; 0.41–0.60, moderate agreement; 0.61–0.80, good agreement; 0.81–1.00, very good agreement (Altman 1991).

Multi-locus sequence analysis (MLSA)

For PCR amplification of specific regions, the partial protein-coding genes used in this study were ADP-ribosylation factor (arf), H-antigen precursor (H-anti), and delta-9 fatty acid desaturase (ole). The reactions were performed as previously described (Kasuga *et al.* 1999) (Table 1). *H. capsulatum* ITS regions were directly amplified from genomic DNA with primers ITS5 and ITS4 (White *et al.* 1990) and used as barcoding markers (Irinyi *et al.* 2015). PCRs were performed in a final volume of 25 µL, including 12.5 µL PCR Master Mix buffer (2×), which consisted of 3 mM $MgCl_2$, 400 mM each dNTP, and 50 U/mL of Taq Polymerase (Promega Corporation, Madison, WI, USA); 9.5 µL of water, 1 µL each of forward and reverse primers (10 pmol/µL; Integrated DNA Technologies, Coralville, IA, USA), and 1 µL of target DNA [100 ng/µL]. The sequencing reactions were carried out with the BigDye Terminator v3.1 Cycle Sequencing Kit (Applied Biosystems, Foster City, CA, USA), and the sequencing products were determined using an ABI3730xL Genetic Analyzer 48-well capillary sequencer (Applied Biosystems). The sequences generated in both orientations were assembled into single sequences via CAP3 implemented in the BioEdit software (Hall 1999). Sequences were aligned with MAFFT v.7 (Katoh & Standley 2013) and the retrieved alignments were manually edited to avoid mispaired bases. All sequences were deposited online with GenBank (accession numbers: MK893472–MK893880; Table S1).

Characterisation of the mating type idiomorphs

PCR primers designed to specifically amplify either the *MAT1-1* or the *MAT1-2* region were used to determine the mating types, as described before (Bubnick & Smulian 2007). Approximately 50 ng of genomic DNA was used for PCR with two sets of oligonucleotide primers: MAT1-1S (5-CGT GGT TAG TTA CGG AGG CA-3) and MAT1-1AS (5-TGA GGA TGC GAG TGA TGG GA-3), which amplify a 440 bp fragment from the α box region of the *MAT1-1* idiomorph, and MAT1-2S (5-ACA CAG TAG CCC AAC CTC TC-3) and MAT1-2AS (5-TCG ACA ATC CCA TCC AAT ACC G-3), which amplify a 528 bp fragment from the HMG domain gene, present in the *MAT1-2* idiomorph (Bubnick & Smulian 2007). PCRs were performed with PCR Master Mix buffer (Promega) as described above under the following conditions: 4 min at 94 °C; followed by 35 cycles of 1 min at 94 °C, 1 min at 60 °C, and 1 min at 72 °C; and a final step of 5 min at 72 °C. Samples were visualised on agarose gels as described above.

Phylogenetic reconstruction

Multi-locus sequence analysis (MLSA) was performed by sequencing PCR-amplified gene fragments. A subset of 104 *H. capsulatum* strains was selected for MLSA (Kasuga *et al.* 2003). The oligonucleotide primer sequences used for MLSA in this study are listed in Table 1. Genetic relationships were investigated by phylogenetic analysis using the neighbor joining (NJ), maximum likelihood (ML), and maximum parsimony (MP) methods. Phylogenetic trees were constructed with MEGA7 (Kumar *et al.* 2016). Evolutionary distances were computed using the Kimura 2-parameter distance (Kimura 1980), and the robustness of branches was assessed by bootstrap analysis of 1 000 replicates (Felsenstein 1985).

Amplified fragment length polymorphism

The amplified fragment length polymorphism (AFLP) analysis was conducted according to a modified version of the protocol (Vos *et al.* 1995) described previously (Najafzadeh *et al.* 2011). The restriction enzymes used were EcoRI (selective primer: 5′-GAC TGC GTA CCA ATT CNN-3′) and MseI (selective primer: 5′-GAT GAG TCC TGA GTA ANN-3′). Selective amplification was carried out with the fluorescent-labeled (FAM) primer pair EcoRI-AC and MseI-CT. The AFLP products were separated with an ABI3100 Genetic Analyzer, along with the GeneScan 500 ROX internal lane size standard (Applied Biosystems). Selection of the amplified restriction products was automated, and only strong and high-quality fragments were considered. The size of the AFLP fragments was determined with the BioNumerics v7.6 software package (Applied Maths). Binary AFLP matrices were created from the presence (1) or absence (0) at probable fragment positions. Pairwise, genetic distances were expressed as the complement of the Dice coefficient (Dice 1945). Dendrograms were produced according to the unweighted pair-group mean arithmetic method (UPGMA). The cophenetic correlation and standard deviation were used to express the consistency of a given cluster, by calculating the correlation between the dendrogram-derived similarities and the matrix similarities. To evaluate the performance of primer pair EcoRI-AC and MseI-CT, the following polymorphism indices for dominant markers were calculated: polymorphic information content (*PIC*) (Botstein *et al.* 1980), expected heterozygosity (*H*) (Liu 1998), effective multiplex ratio (*E*) (Powell *et al.* 1996), arithmetic mean heterozygosity (H_{avp}) (Powell *et al.* 1996), marker index (*MI*) (Powell *et al.* 1996, Varshney *et al.* 2007), discriminating power (*D*) (Tessier *et al.* 1999), and resolving power (*Rp*) (Prevost & Wilkinson 1999).

Dimensioning analysis

Principal component analysis (PCA) and multi-dimensional scaling (MDS) were used as alternative grouping methods, producing three-dimensional plots in which the entries were spread according to their relatedness. Dimensioning techniques were executed with AFLP data after conversion into a band matching table. Automated band matching was performed on all fingerprint entries within the comparison, considering a minimum profiling of 5 %, with the optimization and position tolerances for selecting bands set to 0.10 %. Default settings were applied for PCA and MDS in BioNumerics v7.6, subtracting the average for characters. In addition, the Self-Organizing Map (SOM), a popular artificial neural network algorithm in the unsupervised learning category, was used to classify AFLP entries in a two-dimensional space (map) according to their likeliness (Kohonen 2001). The Kohonen map size was set to 100 (i.e., the number of nodes of the neural network in each direction). All Figs were exported and treated using Corel Draw X8.

Genetic population analysis

The nucleotide (π) as well as the haplotype (Hd) diversities (Nei 1987) were estimated using DnaSP version 6 (Rozas *et al.* 2017). Sites containing gaps and missing data were not considered in the analysis. Haplotype network analysis was constructed using the median-joining method (Bandelt *et al.* 1999), implemented in NETWORK v4.6.1.0 (Fluxus Technology, Suffolk, UK). In addition, recombination possibilities were investigated using the neighbor-net method (Bryant & Moulton 2004), which in the presence of recombination, leads to reticulated relationships, described by the uncorrected-P distance or by the split decomposition method (Bandelt & Dress 1992) both implemented in SplitsTree v.4b06 (Huson & Bryant 2006). Additional measures of recombination were estimated using the PHI-test ($P < 0.05$, demonstrating significant evidence of recombination). Analysis of AFLP data in STRUCTURE (v2.3.4) (Pritchard *et al.* 2000) was performed using the admixture model, allowing alpha to be inferred and assuming correlated allele frequencies, using a burn-in period of 10 000 Markov chain Monte Carlo (MCMC) replications followed by 10 000 sampling replications, with 20 independent runs performed for *K* values one to twelve. Analysis of three-gene multi locus concatenated sequences was performed using the linked model (specifying linkage within individual loci) with a burn-in of 20 000 MCMC replications (after initial 500 admixture replications), followed by 100 000 MCMC sampling replications, with 20 independent runs performed for *K* values one to fifteen. The optimal numbers of burn-in and sampling replications were determined by plotting alpha and Ln-likelihood from extended runs for convergence of the Markov chains. All data were analyzed using the method of Evanno and colleagues as implemented in StructureHARVESTER(v0.6.94) (Evanno *et al.* 2005, Earl & vonHoldt 2012) to determine the optimal number of clusters (*K*). Consensus population distributions were obtained with CLUMPP (v1.1.2) (Jakobsson & Rosenberg 2007), using the full search for the AFLP data, and the greedy algorithm (1 000 000 random repeats) for the multi-locus gene data. Final plots were generated using ggplot2 (Gómez-Rubio 2017) in R (Team 2014). In addition, to assess the existence of topological congruence between dendrograms (AFLP and MLSA) and the associated confidence level, we calculated the congruence index (I*cong*), as described by de Vienne and colleagues (de Vienne *et al.* 2007), based on maximum agreement subtrees (MAST).

RESULTS

To confirm the identity of our 104 Brazilian isolates as *Histoplasma* by PCR, we initially chose three *Histoplasma*-specific primer pairs targeting the gene encoding the M-antigen (Guedes *et al.* 2003) and two SCAR markers (Frías De León *et al.* 2012)

(Fig. 1A). Among the available strains, 95 (91.3 %), 102 (98.1 %) and 102 (98.1 %) samples were positive for Msp2, PCR220 and PCR230 respectively (see Table S1). As a control, positive amplifications for all 104 strains using ITS5 and ITS4 primers targeting the rDNA suggested the absence of PCR inhibitors for isolates that failed to amplify using Msp2 or SCARs primers. The sensitivity levels of Msp2, PCR220 and PCR230 were 91.4 %, 98.1 % and 98.1 %, respectively, and the specificity was 100 % for each assay. Positive predictive values were 100 % (95 % CI 96.2–100) for Msp2 and 100 % (95 % CI 96.4–100) for SCARs markers; and negative predictive values were 62.5 % (95 % CI 40.6–81.7) for Msp2 and 88.3 % (95 % CI 63.6–98.5) for SCARs markers (Table 2). The concordance of results by the *Kappa* measure of agreement was 0.67782 (95 % CI: 0.50300–0.85264; good agreement) between Msp2 and each of the SCARs markers, and 0.93137 (95 % CI: 0.83718–1; very good agreement) between SCARs markers. PCR-based identification was assumed to have powerful diagnostic value with high AUC for Msp2 0.957 (95 % CI: 0.903–0.986; $P < 0.0001$) and SCARs markers 0.990 (95 % CI: 0.952–1; $P < 0.0001$), indicating a low number of false-negatives and no false-positives (Fig. 1B, Table 2). Therefore, these markers were able to recognise several genotypes in the *H. capsulatum* clade.

To determine the validity of the rDNA internal transcribed spacer (ITS) region as a marker for diagnostics of human-pathogenic *Histoplasma* species, we selected 215 ITS sequences (103 generated in this study and 112 reference sequences retrieved from GenBank) for phylogenetic analysis. From these entries, only a few sequences were long enough (>500 bp including ITS1+5.8S+ITS2) to be used for reliable alignment. We covered an epidemiologically diverse panel of clinical isolates with a global distribution including the main endemic areas in the Americas (Argentina, Brazil, Colombia, Cuba, Mexico, Panama and USA), as well as isolates that originated from Belgium, China, Egypt, France, Germany, India, Japan and Malaysia (Fig. 2C). Aligned ITS sequences were 582 bp long, including 497 invariable characters, 81 variable characters, 51 parsimony informative (8.76 %), and 30 singletons. Positions containing gaps and missing data were eliminated. Phylogenetic relationships among 215 sequences representing main genetic groups were inferred based on the Tamura 3-parameter (Gamma) substitution model (Table 3). As the trees obtained from the NJ, ML and MP analyses were identical in their topologies, only the NJ tree with bootstrap support values is presented here (Fig. 2). Using ITS as a single barcode, we were able to identify the *H. capsulatum* complex, but we failed to distinguish among cryptic species or new genotypes, as phylogenetic species were hitchhiking through different clades (e.g., LAm A isolates, recently named *H. suramericanum*).

After a survey in the GenBank to recover protein-coding sequences from *Histoplasma* representing globally distributed isolates, we obtained arf (n = 217), H-anti (n = 195), and ole (n = 193) alignments of 453, 385, and 399 positions, of which 60 (13.2 %), 71 (18.4 %) and 62 (15.5 %) were variable, respectively. Using MEGA7, models with the lowest Bayesian information criterion (BIC) scores were considered to describe the best substitution model and K2+G was selected for arf and H-anti, K2 for ole, and K2+G+I for the concatenated 3-loci dataset (n = 101). These models were used in the NJ and ML analyses (Table 3). Widespread topology differences among arf (Fig. 3), H-anti (Fig. 4), and ole (Fig. 5) were observed, indicating that each independent dataset does not contain enough polymorphisms to resolve all relationships. On the other hand, MLSA greatly improved tree resolution, as the accumulation of nucleotide

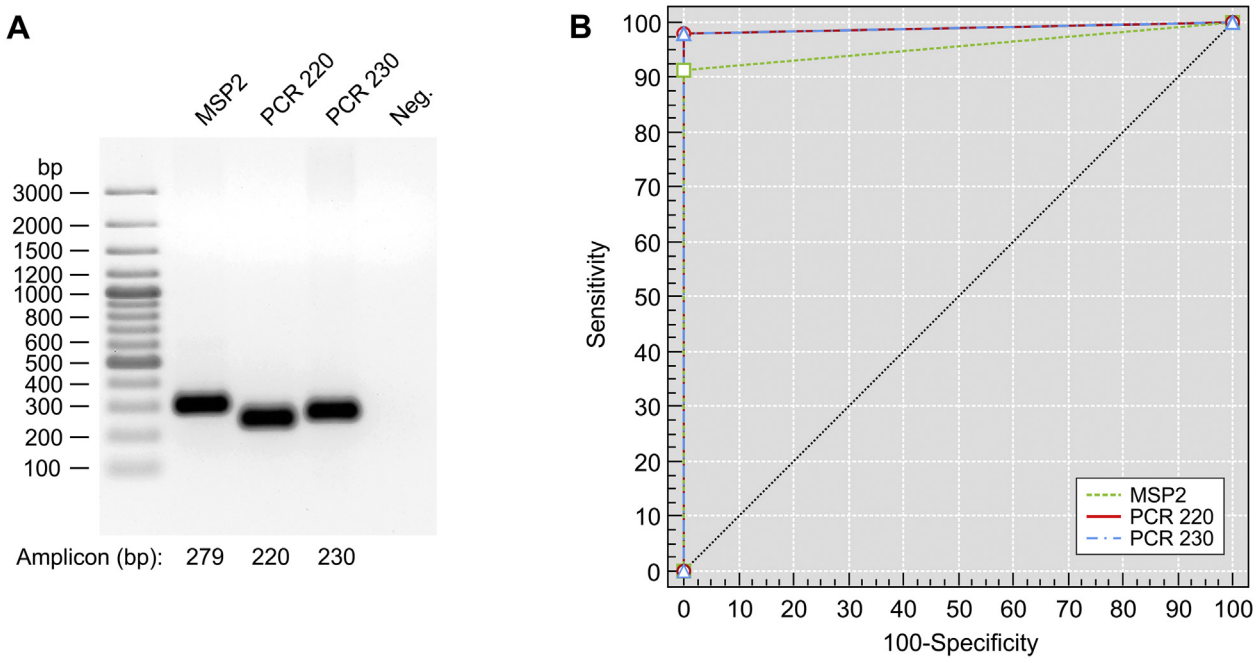


Fig. 1. Molecular identification of *Histoplasma capsulatum* by conventional PCR. **A**. Successful amplification with specific primer sets and *Histoplasma capsulatum* isolate CEMM 05-2-035 (=H2) as a template. Lane 1, 100 bp DNA ladder (Fermentas, USA) for size determinations; Lane 2, Msp2F-Msp2R primer pair amplification (279 bp); Lane 3, 1281-1283$_{220}$ primer pair amplification (PCR 220; 220 bp); Lane 4, 1281-1283$_{230}$ primer pair amplification (PCR 230; 230 bp); Lane 5, negative control. Further information about isolates used and amplification success can be found in Table S1. **B**. Receiver operating characteristic (ROC) curves for primer pairs Msp2F-Msp2R (MSP2), 1281-1283$_{220}$ (PCR 220) and 1281-1283$_{230}$ (PCR 230) based on 104 specimens. Despite high genetic diversity across *H. capsulatum* clusters, primer pairs usually employed in diagnosis successfully identified the new genetic groups recognised in this study. In PCR220 and PCR230, lines are superimposed, indicating equivalent accuracy. The classification variable was a dichotomous variable that indicated the diagnosis (0 = negative, 1 = positive).

Table 2. Comparison of sensitivity, specificity, positive and negative predictive values, and area under the ROC curve among different diagnostic markers used to identify *Histoplasma* spp. cultures (n = 104).

Diagnostic markers	Sen (%)	Sp (%)	PPV (%)	PPV 95 % CI	NPV (%)	NPV 95 % CI	AUC	AUC 95 % CI	AUC *P*-value
M-anti	91.4	100	100	96.2–100	62.5	40.61–81.7	0.957 ± 0.0139	0.903–0.986	<0.0001
PCR220	98.1	100	100	96.4–100	88.3	63.6–98.5	0.990 ± 0.00677	0.952–1	<0.0001
PCR230	98.1	100	100	96.4–100	88.3	63.6–98.5	0.990 ± 0.00677	0.952–1	<0.0001

Sen, sensitivity; Sp, specificity; PPV, positive predictive value; NPV, negative predictive value; AUC, area under the curve.

substitutions produced long branches revealing several cryptic clusters (Fig. 6), here named LAm C-E. Interestingly, genetic clusters often correlate with geographic origin of the isolates, as highlighted by the existence of LAm C, an emerging clade recognised here as causing histoplasmosis in HIV-infected patients who live in Northeast Brazil (Ceará state). In addition, two samples of veterinarian origin (H77 = CEMM 03-3-055 and H78 = CEMM 03-6-059; LAm C) were found as the etiological agent of feline histoplasmosis.

For each *locus* (ITS, arf, ole or H-anti) and for the combined dataset (arf, ole and H-anti), median-joining haplotype networks were built (Figs 2B, 3B, 4B, 5B and 6B), and the Brazilian isolates were analyzed together with the global *Histoplasma* isolates, displaying an overall star-like topology. The network topology clearly reflects the genetic differences among the different clusters observed in MLSA. A low number of median vectors were noted in all networks, especially among protein-coding loci. It often predicts the existence of unsampled or extinct haplotypes, supporting the success of the sampling approach adopted here. Judging from the newly detected group LAm C, there is emergence of a single high frequency haplotype at the center of the network (e.g., Hap 6 in Fig. 2; Hap 3 in Figs 3 and 5; Hap 2 in Fig. 4 and Hap 4 in Fig. 6), surrounded by low frequency derived haplotypes, a pattern typically indicative of recent population expansion or a recent selective sweep (Sarma *et al.* 2012). Nucleotide and haplotype diversities were high for all markers, with a range from π = 0.01080 to 0.01665 and Hd = 0.83 to 0.91; and the ITS dataset with 219 sequences showed the largest number of haplotypes (n = 63), whereas H-anti (Hd = 0.9165) showed higher values of haplotype diversity, followed closely by ITS (Hd = 0.9146) (Table 3). Haplotype networks constructed for different loci of the Brazilian isolates showed ancestral relationships with different cryptic lineages identified from the South American population. Brazilian haplotypes classified as LAm C always showed closer relationship to other sympatric genetic groups and the nearest taxa were LAm E and RJ, followed by LAm A, a clade with isolates of human and animal origin recovered from Brazil, Colombia, Mexico and USA (Fig. 7).

To provide a more accurate representation of genetic groups' relationships, a phylogenetic network was constructed based on the concatenated alignment of the three coding regions by the split-decomposition method, which revealed clear evidence of phylogenic incongruence and potential recombination (Fig. 7A). Remarkably, large conflicting signals were detected, supporting recombination events at some point in evolutionary history. Not surprisingly, the PHI-test strongly supported the occurrence of recombination (*P* = 1.539e-7). STRUCTURE analyses using the MLSA dataset revealed three genetic clusters (*k* = 3) among 179 *Histoplasma* isolates (Fig. 7B). Within each country, the genetic composition of the individual clusters was very heterogeneous, ranging from isolates presenting high membership, especially in LAm C (Northeast Brazil), as well as strains with admixed ancestry, as in LAm D (Northeast, Midwest and Southeast Brazil). Judging from a threshold value used of 0.8 to identify single clusters, all groups presented admixed ancestry (Fig. 7B). Therefore, admixture was identified within the Brazilian populations (Northeast and Southeast), further suggesting that recombination is present among the loci assessed (Fig. 7B).

From a subset of 80 Brazilian isolates, listed in Fig. 8, we obtained 75 variable AFLP markers that could be scored unambiguously as present or absent in each *Histoplasma* isolate (Fig. 8). At a cutoff level of 53 %, the isolates grouped into five clearly separated groups (LAm B–E and RJ; global cophenetic correlation = 0.84; Shannon-Wiener diversity index = 1.05; Simpson's diversity = 52.97 %). The largest group, corresponding to LAm C (n = 53, 66.2 %; cophenetic correlation = 0.64), reveals the emerging genotypes mainly found in Northeast Brazil. Regarding the distribution of the 75 polymorphic markers, we detected the largest number of bands (scored bands = 70; *PIC* = 0.4711) and the presence of 64 markers in at least two isolates clustered in group LAm C, indicating the importance of this group for genetic diversity of *Histoplasma* in Northeast Brazil. Remaining isolates corresponded to the genetic clusters LAm B (n = 13, 16.2 %; cophenetic correlation = 0.88; *PIC* = 0.3164), LAm D (n = 4, 5 %; cophenetic correlation = 1.0; *PIC* = 0.2347), LAm E (n = 3, 3.7 %; cophenetic correlation = 0.99; *PIC* = 0.2392) and RJ (n = 7, 8.7 %; cophenetic correlation = 0.94; *PIC* = 0.2880). We used the allele frequencies to calculate polymorphism indices for dominant markers including the polymorphic information content (*PIC* = 0.3671), expected heterozygosity (*H* = 0.4845), and arithmetic mean heterozygosity (H_{avp} = 0.0001) to reveal that there was remarkable variability among the genetic groups. The informativeness of AFLP markers was also estimated using effective multiplex ratio (*E* = 44.1125), marker index (*MI* = 0.0036), discriminating power (*D* = 0.6541), and resolving power (*Rp* = 30.0750). A strong and positive linear relationship was observed between *PIC* and *RP* values of EcoRI-AC/MseI-CT supporting its ability to distinguish genotypes (Pearson correlation = 0.96, r^2 = 0.93, *P* = 0.0079). A summary of AFLP marker attributes calculated for selective primers EcoRI-AC/MseI-CT of *Histoplasma* species are given in Table 4.

We found strong correlation among genetic diversity, population structure and their geographical origin. STRUCTURE analyses based on AFLP data revealed two populations, one from the Northeast (Population 1 = LAm C, mainly derived from Fortaleza, Ceará) and one from the Southeast (Population 2 = LAm B, LAm D, LAm E and RJ, originated from endemic

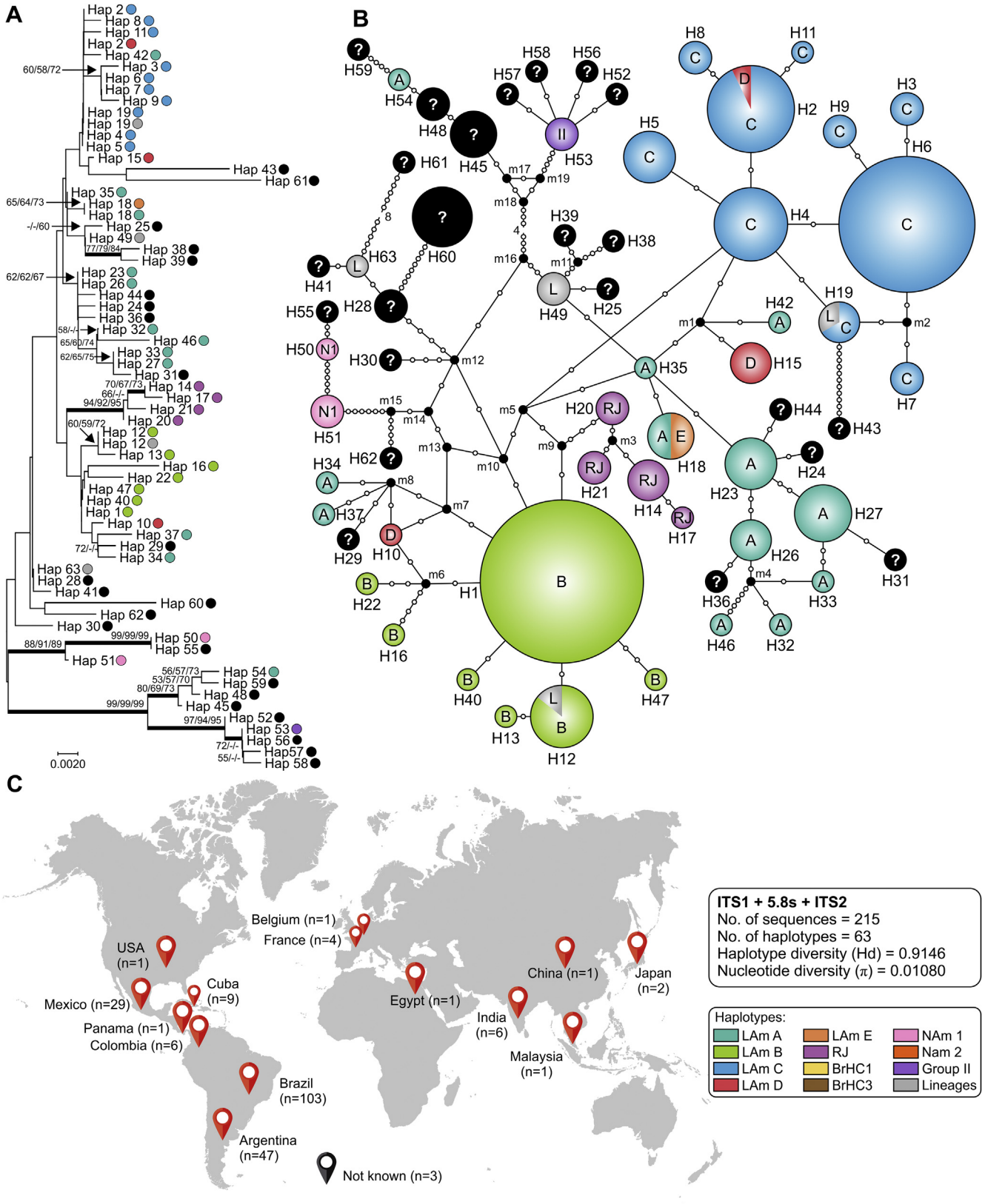


Fig. 2. Phylogeny, haplotype and structure among *Histoplasma capsulatum* genotypes. **A**. Phylogenetic relationships, as inferred from neighbor joining analysis of the ITS sequences (ITS1+5.8S+ITS2; n = 215 OTU), covering the main haplotypes of *H. capsulatum*. Numbers above the tree branches are the bootstrap values for NJ, ML and MP methods. The branches with bootstrap support higher than 70 % are indicated in bold. **B**. Median-joining haplotype network of *H. capsulatum* isolates, covering all the ITS haplotypes found in this study. The size of the circumference is proportional to the haplotype frequency. The haplotypes are color coded according to the genetic group to which they were assigned. Mutational steps are represented by white dots. The black dots (median vectors) represent unsampled or extinct haplotypes in the population. **C.** Distribution patterns of *H. capsulatum* ITS sequences used in this study. Note that clade naming follows the original appearance of the isolates in the literature (Kasuga *et al.* 2003, Taylor *et al.* 2005, Muniz Mde *et al.* 2010). NJ, neighbor joining; ML, maximum likelihood; MP, maximum parsimony. Further information about isolate source and GenBank accession number can be found in Table S1.

Table 3. Amplification success, phylogenetic data and the substitution models used in the phylogenetic and haplotype analysis, per locus.

Locus/Region		ARF	H-anti	OLE	ITS	MLSA
Amplification success (%)		97.11	99.03	98.07	99.03	–
Histoplasma isolates	This study (n)	101	103	102	103	101
	GenBank (n)	116	92	91	116	77
	Total (n)	217	195	193	219	178
Best model		K2+G	K2+G	K2	T92+G	K2+G+I
MP–consistency index		0,814815	0,704918	0,750000	0,661765	0,500000
MP–retention index		0,960317	0,920000	0,921986	0,910156	0,901961
MP–composite index (for all sites)		0,837200	0,749278	0,786761	0,714515	0,572341
MP–Tree length		78	97	75	107	301
MP–No. of tress		9	2	1	5	2
N. of sequences		211	188	192	215	179
N. of sites		453	385	399	582	1237
Total number of sites (excluding sites with gaps / missing data):		430	369	390	550	1192
Conserved characters		392	309	336	497	1053
Variable characters		60	71	62	81	176
Parsimony-informative characters		40	38	48	51	98
Singletons		20	33	14	30	78
Nucleotide diversity (pi)		0,01397	0,01665	0,01305	0,01080	0,01346
Number of haplotypes (h)		55	52	44	63	88
Haplotype diversity (Hd)		0,9101	0,9165	0,8360	0,9146	0,9746
Recombination phi test (Φ) (*P*-value)		No (P = 0.3754)	No (P = 0.1068)	No (P = 0.6897)	No (P = 0.3)	Yes (P = 1.539e-7)

ARF: ADP-ribosylation factor; H-anti: H antigen precursor; OLE: Delta-9 fatty acid desaturase; ITS: Internal transcribed spacers + rRNA genes; MLSA: Multi-locus sequence analysis; MP: maximum parsimony.

areas in the states of Rio de Janeiro, São Paulo and Espírito Santo). In addition, the population structure reflects shared AFLP markers among individuals embedded in cluster LAm E, which shared 54 markers (84.3 %) with LAm C and 27 markers with LAm D. Therefore, our data support LAm D (E = 45.2500; *Havp* = 0.0013) and LAm E (E = 45.0000; *Havp* = 0.0017) as hybrid groups in *Histoplasma*. On the other hand, a low degree of shared markers was observed among the LAm C and the clusters LAm B (E = 43.0769; *Havp* = 0.0005) and RJ (E = 51.1429; *Havp* = 0.0008), suggesting longstanding separation, as recently suggested for other cryptic species in *Histoplasma* (Teixeira *et al.* 2016, Sepúlveda *et al.* 2017).

Our AFLP data support the view that cryptic speciation is a common phenomenon in *Histoplasma* and that cryptic species can be recognised using fingerprint markers. Furthermore, this study shows that AFLP data are a valuable supplement to DNA sequence data in that they may detect a finer level of genetic variation. In this scenario, the topologies of MLSA trees and the AFLP dendrogram showed strict correspondence, with all clinical clades being recognised using both markers (P = 2.28 e-05; I*cong* = 1.52) (de Vienne *et al.* 2007). However, single loci analysis based on protein coding regions or ITS region as a barcoding marker failed to recognise cryptic species in separate analyses.

Analysis of AFLP data through PCA (Fig. 9A) and MDS (Fig. 9B) corroborates the clusters recognised by MLSA analysis. PCA revealed that the first three principal components described 35.9 % of the variation in the AFLP data (Fig. 9). Self-organizing maps based on AFLP fingerprints were used and are plotted according to genetic clusters (Fig. 9C) or AFLP populations (Fig. 9D). Unlike PCA, the distance between entries in the SOMs is not in proportion to the taxonomic distance between the entries. Rather, SOMs contain areas of high distance and areas of high similarity (Felix *et al.* 2015). In Fig. 9C,D, the lighter and thicker the line (white, gray) between black blocks, the more distant are those samples contained in the black block, from the adjacent black block. The models were organized into a meaningful two-dimensional order in which similar *Histoplasma* genotypes are closer to each other in the grid than the more dissimilar ones. Closely related LAm C genotypes are bounded by thin faint white/gray lines (e.g., isolates H44, H46, H48 and H49 from LAm C, Population 1; all belonging to MLSA Hap 04), implying high similarity of the strains, supporting the emergence of a group with low genetic diversity among HIV patients (Fig. 9C). On the other hand, genetically distant samples are separated by lighter and thicker lines (white, gray) between black blocks (e.g., isolates belonging to population 1 versus population 2; Fig. 9D).

A mating type-specific PCR assay was used to amplify either the *MAT1-1* or the *MAT1-2* region among 104 *H. capsulatum* isolates. The *MAT1-1* region was observed in 54 isolates, while the *MAT1-2* region was observed among 49 isolates. The only exception was observed for isolate EPM1011 (LAm B, São Paulo, Brazil), which failed to amplify both *MAT* alleles despite positive amplification using ITS5 and ITS4 primers, therefore excluding the possibility of PCR inhibitors. The distribution of each sexual idiomorph within a geographical region (Northeast and Southeast) and molecular types (LAm B-E, RJ) are presented in Table 5. The distributions of *MAT1-1* or *MAT1-2* idiomorph were not significantly skewed (1:1 ratio) within Brazilian states or molecular types, as

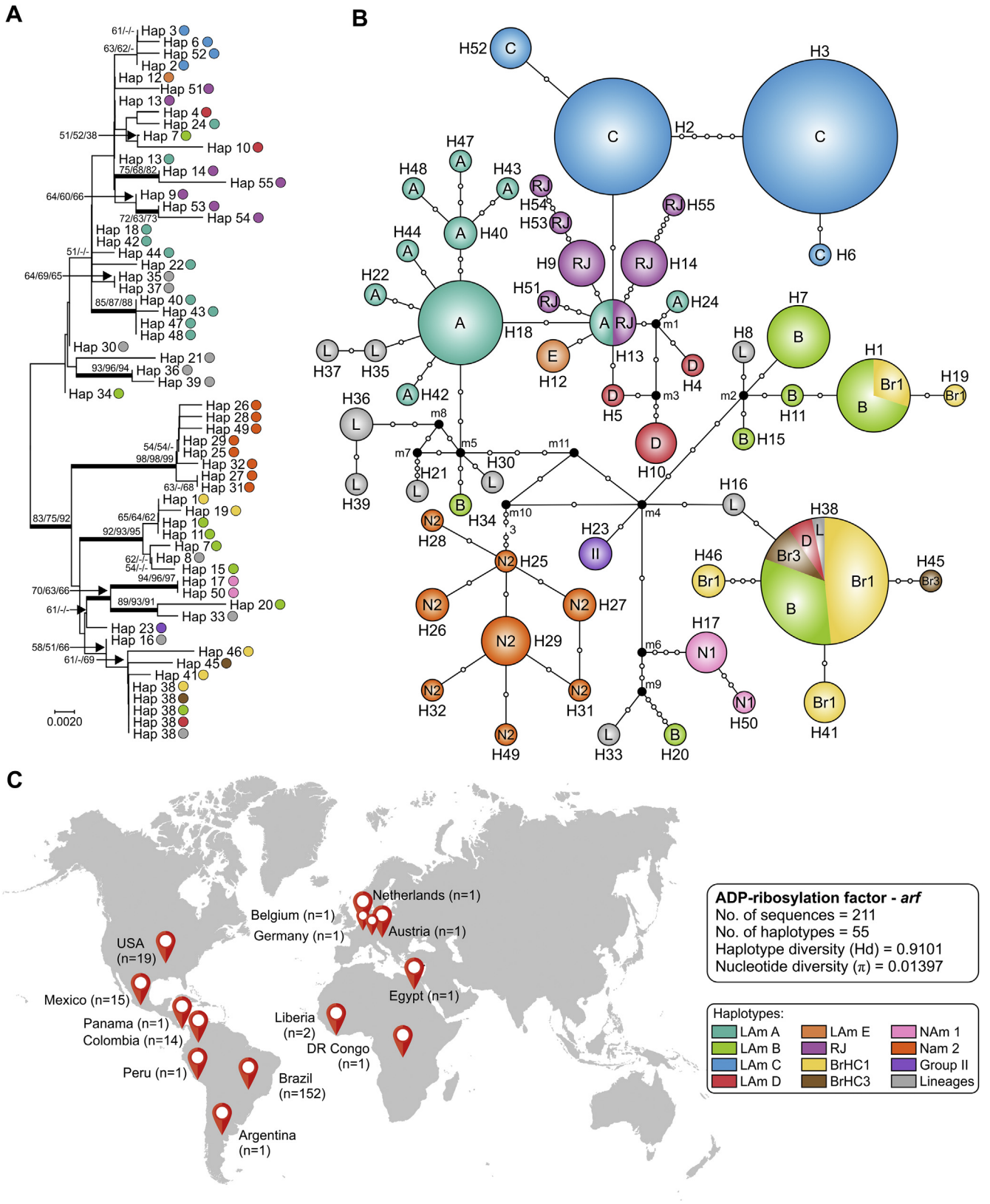


Fig. 3. Phylogeny, haplotype and structure among *Histoplasma capsulatum* genotypes. **A**. Phylogenetic relationships, as inferred from neighbor joining analysis of the ADP-ribosylation factor sequences (arf; n = 211 OTU), covering the main haplotypes of *H. capsulatum*. Numbers above the tree branches are the bootstrap values for NJ, ML and MP methods. The branches with bootstrap support higher than 70 % are indicated in bold. **B**. Median-joining haplotype network of *H. capsulatum* isolates, covering all the arf haplotypes found in this study. The size of the circumference is proportional to the haplotype frequency. The haplotypes are color coded according to the genetic group to which they were assigned. Mutational steps are represented by white dots. The black dots (median vectors) represent unsampled or extinct haplotypes in the population. **C**. Distribution patterns of *H. capsulatum* arf sequences used in this study. Note that clade naming follows the original appearance of the isolates in the literature (Kasuga *et al.* 2003, Taylor *et al.* 2005, Muniz Mde *et al.* 2010). NJ, neighbor joining; ML, maximum likelihood; MP, maximum parsimony. Further information about isolate source and GenBank accession number can be found in Table S1.

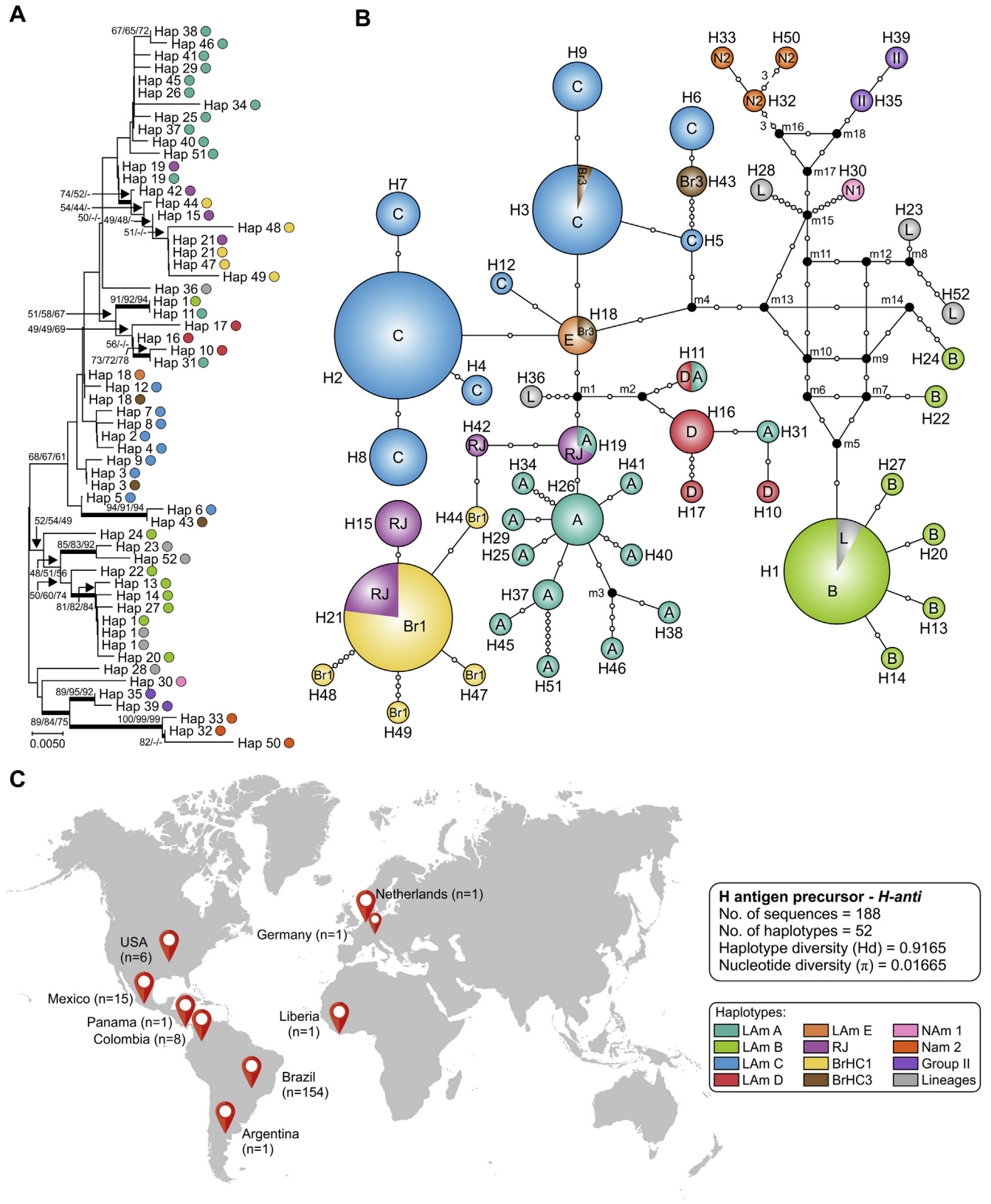


Fig. 4. Phylogeny, haplotype and structure among *Histoplasma capsulatum* genotypes. **A**. Phylogenetic relationships, as inferred from neighbor joining analysis of the H antigen precursor sequences (H-anti; n = 188 OTU), covering the main haplotypes of *H. capsulatum*. Numbers above the tree branches are the bootstrap values for NJ, ML and MP methods. The branches with bootstrap support higher than 70 % are indicated in bold. **B**. Median-joining haplotype network of *H. capsulatum* isolates, covering all the H-anti haplotypes found in this study. The size of the circumference is proportional to the haplotype frequency. The haplotypes are color coded according to the genetic group to which they were assigned. Mutational steps are represented by white dots. The black dots (median vectors) represent unsampled or extinct haplotypes in the population. **C**. Distribution patterns of *H. capsulatum* H-anti sequences used in this study. Note that clade naming follows the original appearance of the isolates in the literature (Kasuga *et al.* 2003, Taylor *et al.* 2005, Muniz Mde *et al.* 2010). NJ, neighbor joining; ML, maximum likelihood; MP, maximum parsimony. Further information about isolate source and GenBank accession number can be found in Table S1.

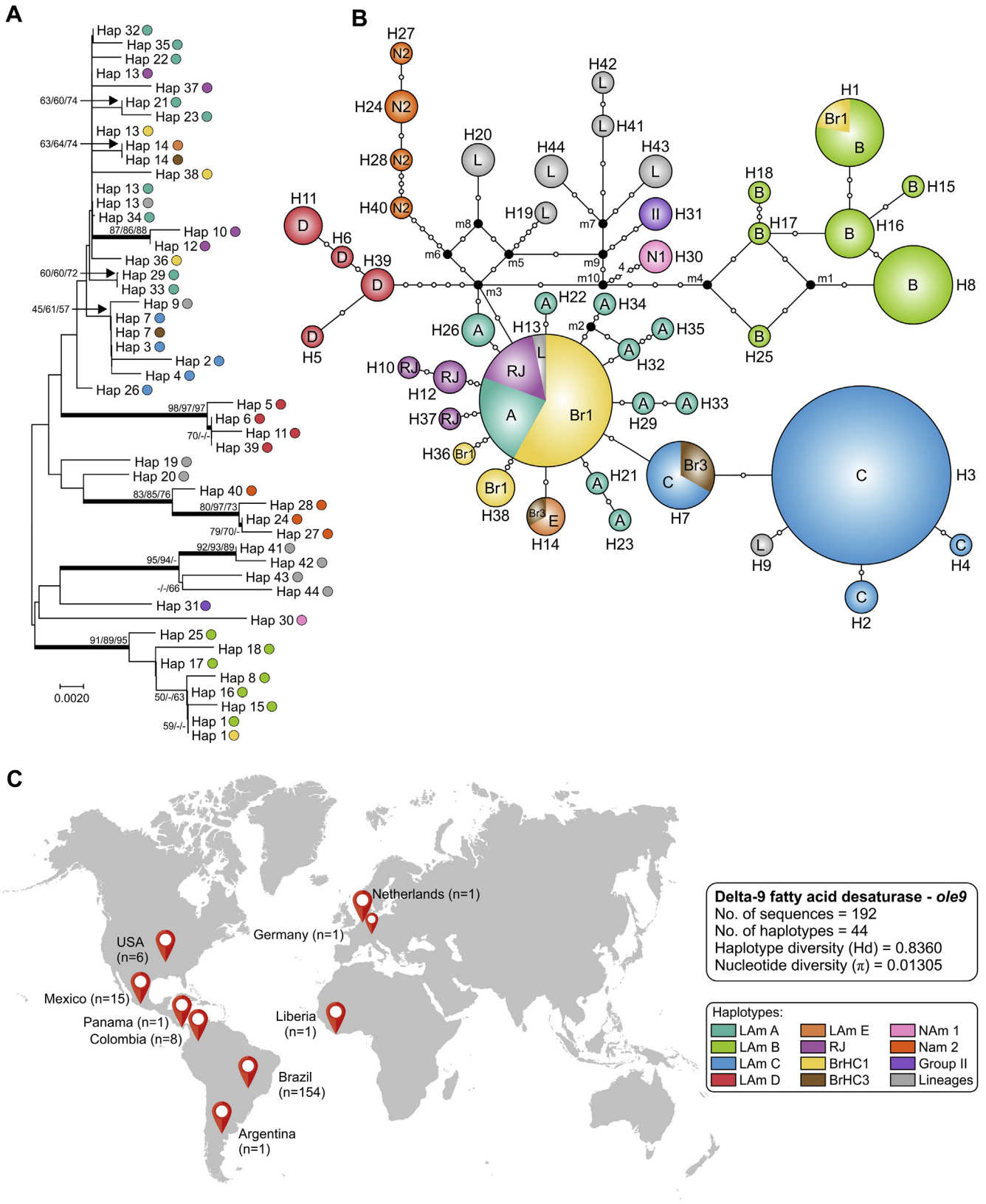


Fig. 5. Phylogeny, haplotype and structure among *Histoplasma capsulatum* genotypes. **A**. Phylogenetic relationships, as inferred from neighbor joining analysis of the delta-9 fatty acid desaturase sequences (ole; n = 192 OTU), covering the main haplotypes of *H. capsulatum*. Numbers above the tree branches are the bootstrap values for NJ, ML and MP methods. The branches with bootstrap support higher than 70 % are indicated in bold. **B**. Median-joining haplotype network of *H. capsulatum* isolates, covering all the ole haplotypes found in this study. The size of the circumference is proportional to the haplotype frequency. The haplotypes are color coded according to the genetic group to which they were assigned. Mutational steps are represented by white dots. The black dots (median vectors) represent unsampled or extinct haplotypes in the population. **C**. Distribution patterns of *H. capsulatum* ole sequences used in this study. Note that clade naming follows the original appearance of the isolates in the literature (Kasuga *et al.* 2003, Taylor *et al.* 2005, Muniz Mde *et al.* 2010). NJ, neighbor joining; ML, maximum likelihood; MP, maximum parsimony. Further information about isolate source and GenBank accession number can be found in Table S1.

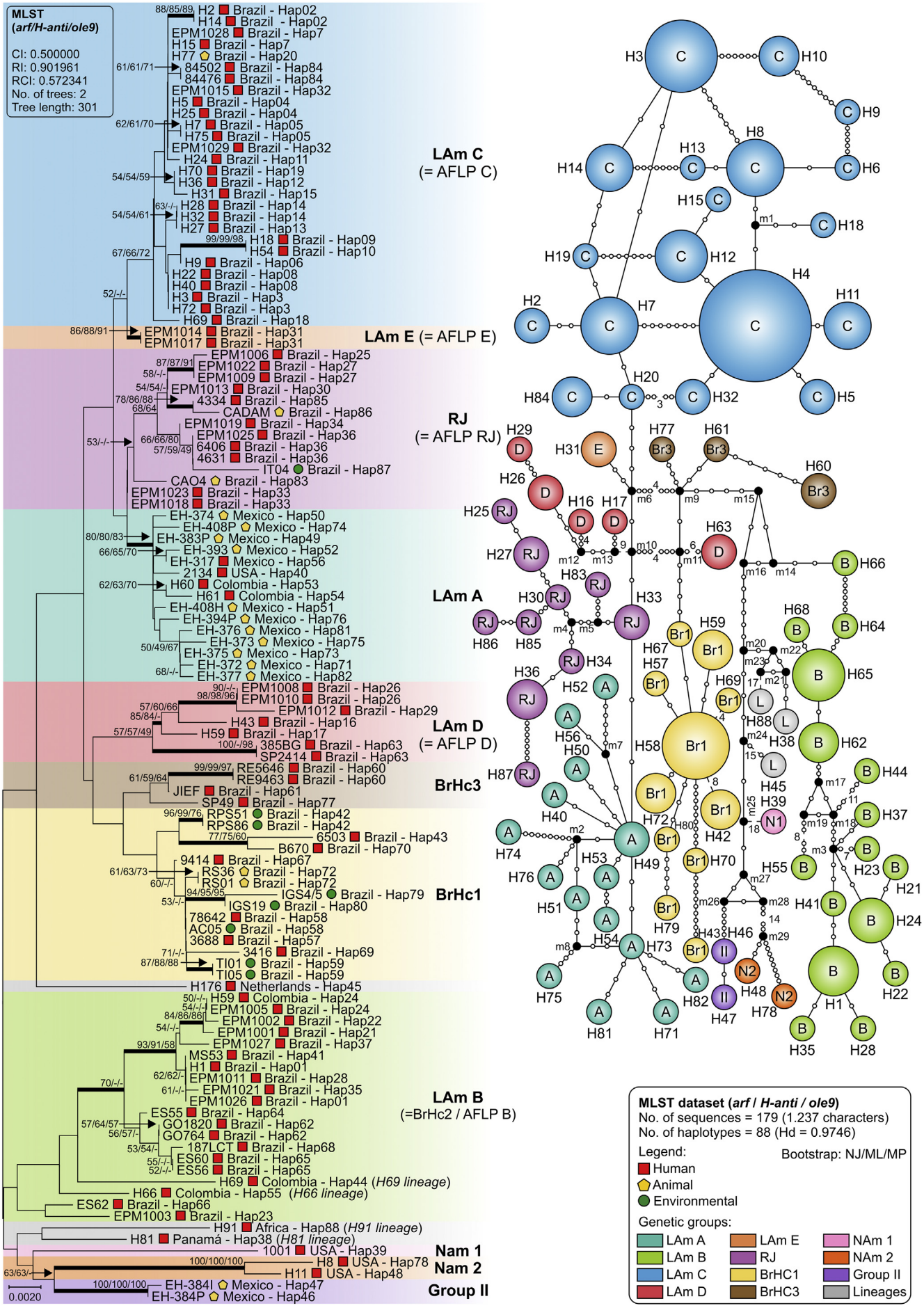


Fig. 6. Phylogeny and haplotype among *Histoplasma capsulatum* genotypes. **A**. Phylogenetic relationships, as inferred from neighbor joining analysis of concatenated ADP-ribosylation factor, H-antigen precursor and delta-9 fatty acid desaturase sequences (n = 179 OTU), covering the main genetic groups of *H. capsulatum*. Numbers above the

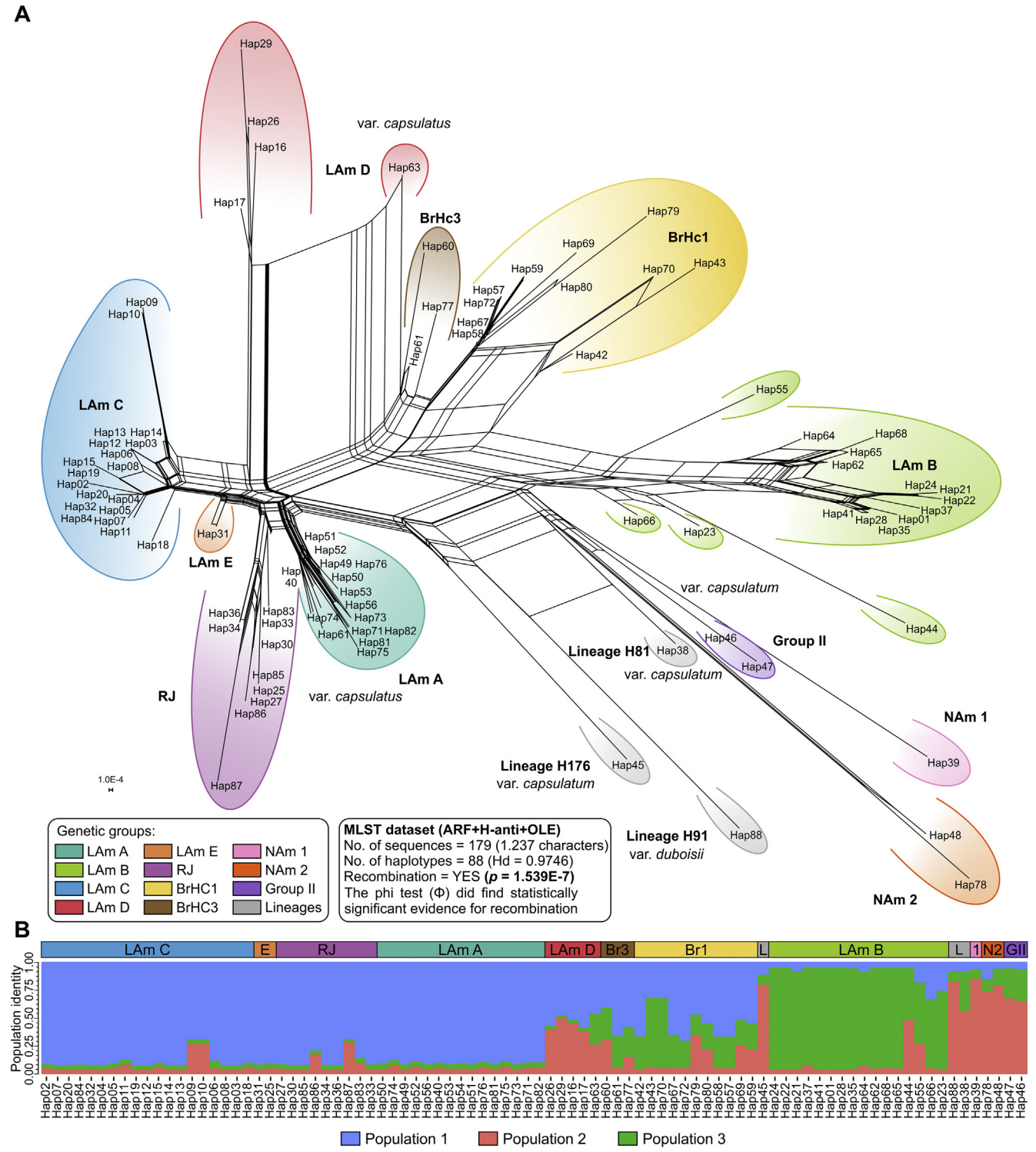


Fig. 7. A. The neighbor network using the uncorrected p-distance among a core set of *Histoplasma* genotypes based on 1 237 nucleotide positions derived from the ADP-ribosylation factor, H-antigen precursor and delta-9 fatty acid desaturase loci (n = 179 OTU). Sets of parallel edges (reticulations) in the networks indicate locations of incongruence and potential recombination. Recombination within *H. capsulatum* genotypes was also supported by the PHI-test (Φ) (*P* = 1.539E-7). **B**. Bayesian cluster analyses with STRUCTURE of 88 *H. capsulatum* haplotypes based on MLSA dataset. Each vertical bar represents one individual MLSA haplotype and its probabilities of being assigned to clusters. Note that clade naming follows the original appearance of the isolates in the literature (Kasuga *et al.* 2003, Taylor *et al.* 2005, Muniz Mde *et al.* 2010). Further information about isolate source can be found in Table S1.

tree branches are the bootstrap values for NJ, ML and MP methods. The branches with bootstrap support higher than 70 % are indicated in bold. **B**. Median-joining haplotype network of *H. capsulatum* isolates. The size of the circumference is proportional to the haplotype frequency. The haplotypes are color coded according to the genetic group to which they were assigned. Mutational steps are represented by white dots. The black dots (median vectors) represent unsampled or extinct haplotypes in the population. Note that clade naming follows the original appearance of the isolates in the literature (Kasuga *et al.* 2003, Taylor *et al.* 2005, Muniz Mde *et al.* 2010). ML, maximum likelihood; MP, maximum parsimony; NJ, neighbor joining. Further information about isolate source and GenBank accession number can be found in Table S1.

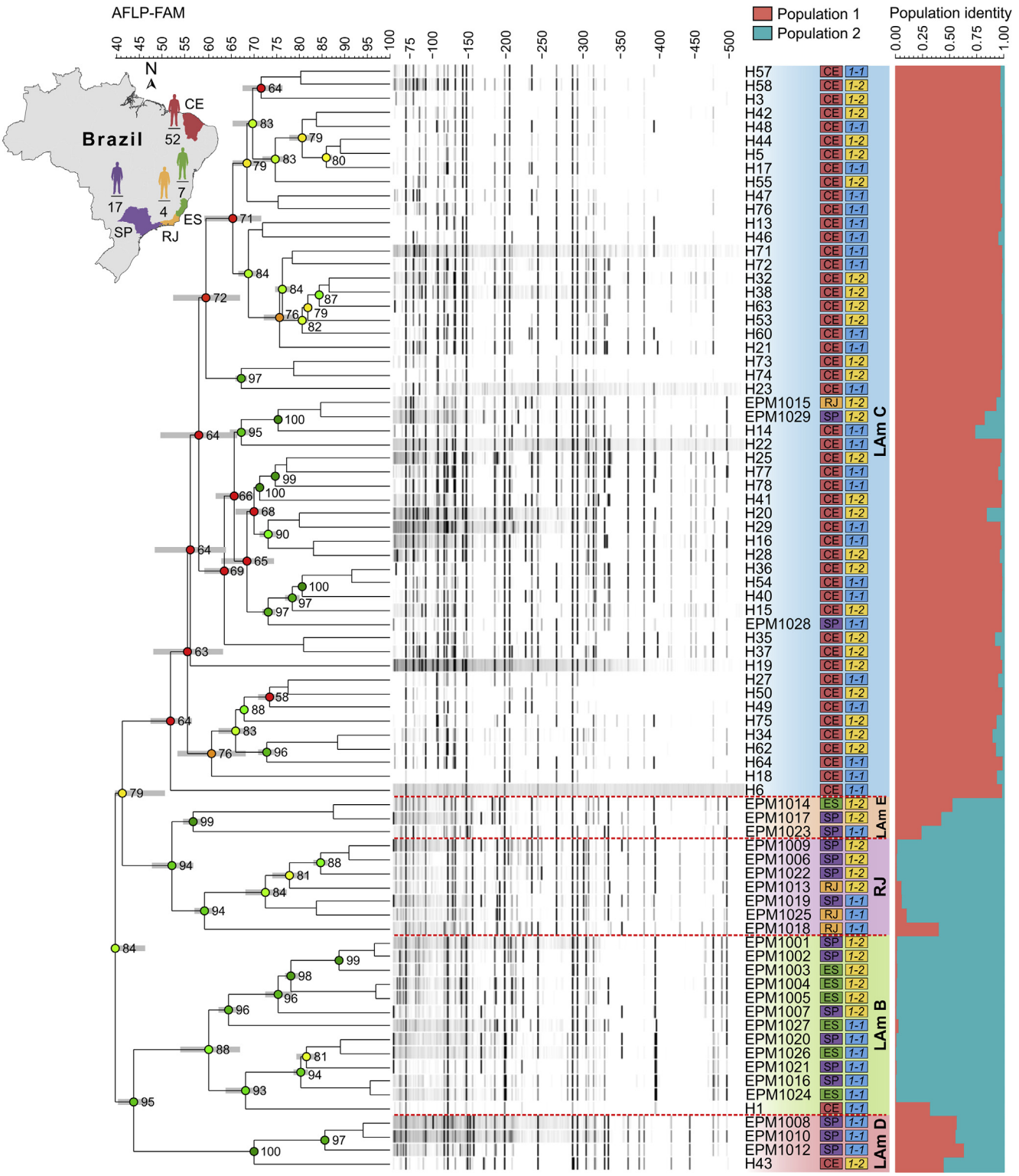


Fig. 8. The UPGMA dendrogram based on amplified fragment length polymorphism (AFLP) fingerprint, generated with a total of four selective bases (EcoRI-AC/MseI-CT) for 80 *Histoplasma capsulatum* originated from Brazil. AFLP results show a significant grouping according to geographical origin (Northeast and Southeast Brazil) and MLSA clusters, revealing five differentiated AFLP groups (LAm B, C, D, E and RJ). The dendrogram shows cophenetic correlation values (circles) for a given clade and its standard deviation (gray bar). Patterns of mating-type idiomorphs' distribution across AFLP groups are shown. Fragments between 50 and 500 bp are shown. For pairwise, genetic distances calculation, Dice coefficient was used. The cophenetic correlation of the dendrogram is 0.84. Bayesian cluster analyses with STRUCTURE (k = 2) of 80 *H. capsulatum* samples based on AFLP. Each vertical bar represents one individual and its probabilities of being assigned to clusters. Further information about isolate source can be found in Table S1. CE, Ceará (Northeast); ES, Espírito Santo; RJ, Rio de Janeiro; SP, São Paulo (Southeast); *1-1*, *MAT1-1* mating type idiomorph; *1-2*, *MAT1-2* mating type idiomorph. Note that clade naming follows the original appearance of the isolates in the literature (Kasuga *et al.* 2003, Taylor *et al.* 2005, Muniz Mde *et al.* 2010).

evidenced by Chi-square tests, supporting the presence of random mating within each population.

Although classically histoplasmosis is described as an endemic disease in the Americas, our data call attention to global distribution of the disease related to different genotypes of *Histoplasma* (Fig. 10). Brazil harbors nine highly diverse genetic groups of *Histoplasma*, and here we describe the recent emergence of LAm C among HIV patients living in Fortaleza, Ceará.

Table 4. Summary of polymorphism statistics calculated for selective primers EcoRI-AC/MseI-CT of *Histoplasma* species.

Genetic group	Scored bands	*H*	*PIC*	*E*	*Havp*	*MI*	*D*	*Rp*
LAm B (n = 13)	59	0.3941	0.3164	43.0769	0.0005	0.0221	0.4672	18.1538
LAm C (n = 53)	70	0.4711	0.3601	43.4151	0.0001	0.0055	0.6154	26.0755
LAm D (n = 4)	54	0.2716	0.2347	45.2500	0.0013	0.0569	0.2984	9.5000
LAm E (n = 3)	54	0.2778	0.2392	45.0000	0.0017	0.0772	0.3064	12.0000
RJ (n = 7)	66	0.3489	0.2880	51.1429	0.0008	0.0386	0.3999	19.1429
Overall (n = 80)	75	0.4845	0.3671	44.1125	0.0001	0.0036	0.6541	30.0750

D = discriminating power; *E* = effective multiplex ratio; *H* = expected heterozygosity; *Havp* = mean heterozygosity; *MI* = marker index; *PIC* = polymorphism information content; *Rp* = resolving power.

Outside Brazil, the predominant genetic group depends on the region. Although most of the cases reported in the literature correspond to the case series observed in the USA (Benedict & Mody 2016, Benedict *et al.* 2020), more specifically in Mississippi and Ohio where *H. mississippiense* and *H. ohiense* are endemic, no taxonomic diversity is observed (Fig. 10).

DISCUSSION

The history of human histoplasmosis can be separated into four periods. The first began in 1906 when Darling described a rare and often fatal disease (Darling 1906). Then during the second period (1940s), most of the sero-epidemiological studies were conducted, employing the histoplasmin skin test, which revealed that Darling's disease was very common and widely distributed, causing pulmonary infection ranging from benign to fatal cases (Christie & Peterson 1945, Joseph Wheat 2003, Antinori 2014). We are now at the end of the third period, where molecular data provide a unique opportunity to recognise and identify phylogenetic species and genetic populations (Kasuga *et al.* 1999, Kasuga *et al.* 2003, Sepúlveda *et al.* 2017). A well-corroborated phylogenetic classification has now begun to emerge for *Histoplasma*. The fourth period will involve comparative genomic studies, based on whole-genome sequencing data, for which purpose a global sampling strategy will be necessary (Sepúlveda *et al.* 2017). A few examples of other taxonomic distinctions of closely related clades in the Onygenales include *Blastomyces* (Brown *et al.* 2013), *Coccidioides* (Fisher *et al.* 2002), *Emergomyces* (Dukik *et al.* 2017), *Emmonsia* (Jiang *et al.* 2018) and *Paracoccidioides* (Turissini *et al.* 2017).

Our study describes new phylogenetic species and the molecular characteristics of *Histoplasma* lineages causing outbreaks with a high number of severe outcomes in Northeast Brazil between 2011 and 2015. A noteworthy difference was observed for the distribution of the emerging genotype LAm C and the affected host population. For instance, *H. capsulatum* LAm C is the major lineage causing histoplasmosis in HIV patients in Northeast Brazil (Ceará), while it shows low genetic diversity, as revealed by the haplotype networks and Kohonen self-organizing maps. Interestingly, this group first appeared in Rio de Janeiro in 1998 (subclade BrHc1C = isolates 84476, 84502, and 84564), but with low frequency (Muniz Mde *et al.* 2010). Moreover, LAm C has not been reported outside Brazil, as observed from sequences deposited in the GenBank, and subsequent re-analysis has revealed a low degree of homology. Similarly, *H. capsulatum* LAm D, E and RJ were recovered only from patients in the Southeast region of Brazil. To some extent, our study confirms previous molecular studies that demonstrated high genetic variability in *H. capsulatum* and structured populations with limited geographic distribution in Brazil (Zancope-Oliveira *et al.* 2005, Muniz Mde *et al.* 2010, Brilhante *et al.* 2012). Muniz and co-workers (Muniz Mde *et al.* 2010) grouped 51 isolates into three major clusters using several DNA fingerprinting methods, which correlates with their geographical origins. BrHc1, with subclades BrHc1A–D, was predominantly found in Rio de Janeiro, followed by Mato Grosso do Sul and São Paulo. Clade BrHc2, a more widespread group spanning an area of over 2 000 km, was detected in Espírito Santo, Rio Grande do Sul, and central Brazil. BrHc3 was found in Pernambuco and Ceará. MLSA revealed that clade BrHc3, including isolates RE5646 and RE9463 from Pernambuco (Northeast Brazil) or BrHc1 (from Rio de Janeiro) are genetically distant taxa from LAm C, supporting the novelty of the emergence described here. This agrees with data from a study reported previously by Teixeira and colleagues (Teixeira *et al.* 2016), who used the MLSA scheme proposed by Kasuga to investigate the global phylogenetic relationship.

From an epidemiological perspective, we present here the largest survey to date (n = 436 isolates; Fig. 10). Our phylogenetic analysis reveals that histoplasmosis is wider spread than previously assumed, indicating that etiological agents can be divided into several phylogenetically distinct groups (Teixeira *et al.* 2016). Our findings go beyond the genetic diversity previously reported, which strengthens the importance of more epidemiological investigations in Brazil. This is supported by the high incidence of histoplasmosis in Brazil, with 2.19 cases per 1 000 hospital admissions (Giacomazzi *et al.* 2016), especially among HIV patients (Almeida *et al.* 2019, Falci *et al.* 2019). The geographic center of origin should contain the highest allelic and genotypic diversities, which was confirmed here by MLSA and AFLP analysis, supporting the hypothesis that Brazil is the center of origin of *Histoplasma* spp. in Latin America, not only due to genetic diversity, but also driven by recombination and evidence of sexual reproduction in these populations. Remarkably, in some pathogenic fungi such as *Aspergillus* and *Cryptococcus*, abundant evidence of sexual recombination has also been used as a signature for an ancestral population (Hagen *et al.* 2013, Ashu *et al.* 2017).

Recent global estimates are of annual incidence of ~100 000 cases of disseminated histoplasmosis (Bongomin *et al.* 2017). However, in many countries the annual incidence of histoplasmosis is not known (Denning 2016), and little or no genetic information has been reported in studies of the agents of histoplasmosis around the world. In Latin America,

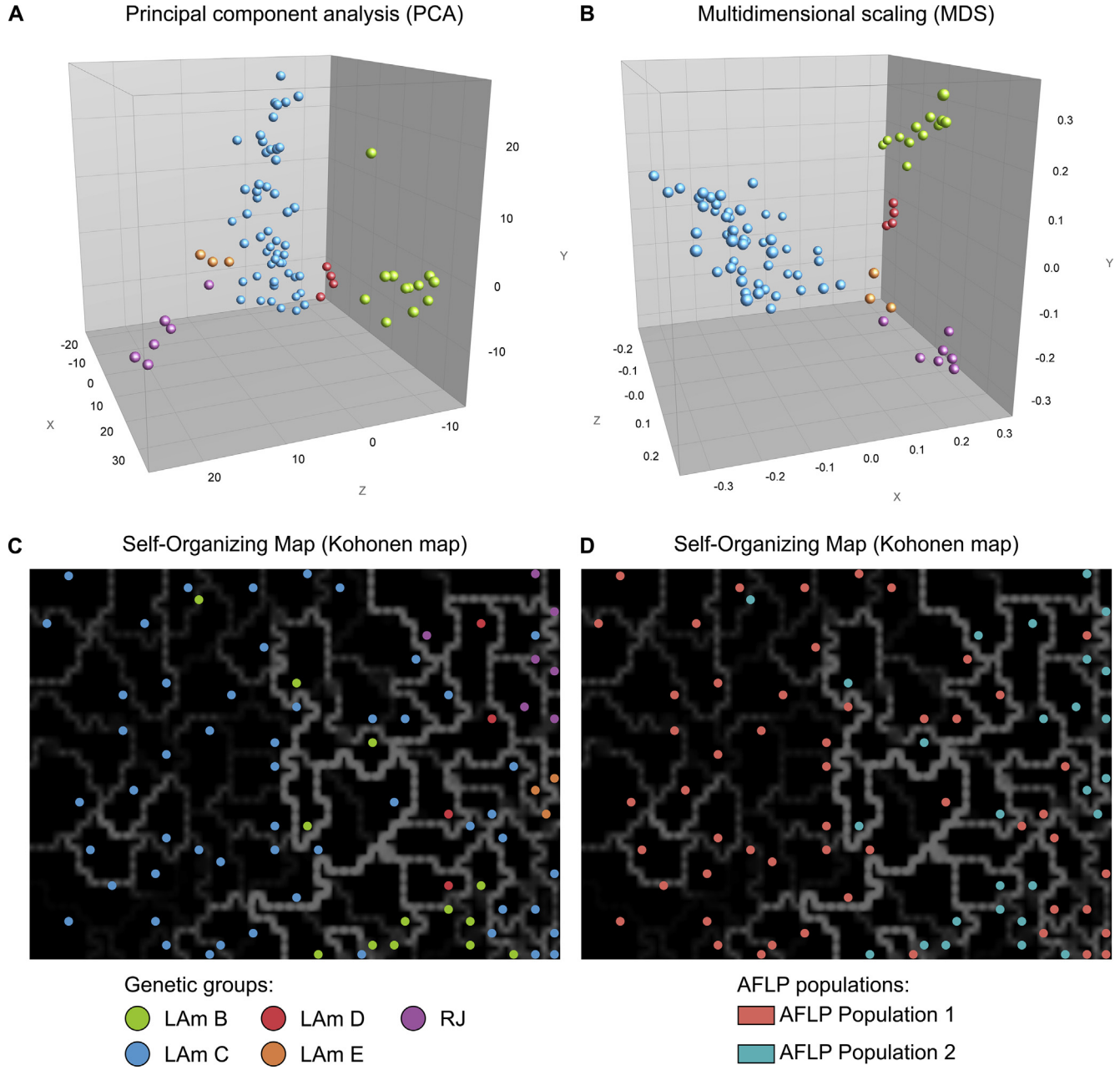


Fig. 9. The distribution of the studied AFLP fingerprints (EcoRI-AC/MseI-CT) of 80 *Histoplasma capsulatum* originated from Brazil, using principal component analysis (PCA), multi-dimensional scaling (MDS) and self-organizing mapping (SOM). The dimensioning analyses were performed using BioNumerics v7.6 to determine the consistency of the differentiation of the populations defined by the cluster analysis. **A** and **B** show the PCA and MDS of AFLP data with the first three principal components describing the greatest variation plotted on the X (15.4 %), Y (12 %), and Z (8.6 %) axes. The SOM revealed that Northeast isolates (n = 50; Ceará) are embedded in areas of high similarity mostly bounded by thin faint white/gray lines according to the genetic group (**C**) or AFLP population (**D**), confirming the low genetic diversity in MLSA data. The lighter and thicker the line (white, gray) between black blocks, the more distant are those samples contained in the black block, from the adjacent black block. Isolates were color coded according to their genetic groups (A, B and C) or AFLP population (D).

for example, histoplasmosis is one of the most common opportunistic infections among people living with HIV, and approximately 30 % of HIV/AIDS patients diagnosed with histoplasmosis die from it, especially among patients who did not receive highly active antiretroviral therapy (HAART) (Colombo *et al.* 2011). Moreover, the pulmonary manifestations may be misdiagnosed as tuberculosis (Linder & Kauffman 2019). Here we provide strong evidence based on molecular epidemiology that there is enormous genetic diversity of *Histoplasma* in Brazil, a situation that deserves to be explored in terms of virulence traits, antifungal susceptibility and clinical outcomes.

Histoplasma capsulatum and allied species cause devastating infection in immunocompromised patients, which is difficult to diagnose rapidly enough to save the patient's life (Nacher *et al.* 2018). When new genotypes were detected in Brazil, we wondered whether the existing molecular assays would be capable of detecting such diversity. Surprisingly, we obtained satisfactory results for detection of *Histoplasma* DNA using Msp2, PCR220 and PCR230 assays, which provided a generic identification as *Histoplasma* spp. Molecular techniques can detect *Histoplasma* spp. even in cases with negative serology (e.g., immunoblotting or double immunodiffusion) negative direct mycological examination (potassium hydroxide wet mount slides) or negative cultures, potentially enabling early diagnosis (Dantas *et al.* 2018).

DNA barcoding indicated that *H. capsulatum s. str., H. mississippiense*, *H. ohiense* and *H. suramericanum* (Sepúlveda

Table 5. Distribution of mating type alleles as determined by PCR with mating type allele-specific primers in *Histoplasma capsulatum* isolates.

Origin/Group	No. of isolates	No. of isolates by mating type		Chi-square value	*P*-value
		***MAT 1-1* (%)**	***MAT 1-2* (%)**		
Histoplasma (Northeast)	75	40 (53,33)	35 (46,66)	0.333	0.5637
Histoplasma (Southeast)	29[1]	14 (50,00)	14 (50,00)	0	1.0000
Histoplasma (Brazil)	104[1]	54 (52,42)	49 (47,57)	0.243	0.6222
LAm B	14[1]	7 (53,84)	6 (46,15)	0.077	0.7815
LAm C	74	40 (54,05)	34 (45,94)	0.486	0.4855
LAm D	5	3 (60,00)	2 (40,00)	0.200	0.6547
LAm E	2	0 (0,00)	2 (100,00)	-	-
RJ	8	4 (50,00)	4 (50,00)	0	1.0000

[1] One isolate (EPM1011; Brazil - Southeast, LAm B) failed to amplify a *MAT* gene product.

et al. 2017), the classic and new phylogenetic cryptic groups (Kasuga *et al.* 2003) and the historical taxonomic varieties (var. *capsulatum*, var. *farciminosum*, and var. *duboisii*) could not be entirely distinguishable by fixed mutations in the ITS1+5.8-S+ITS2 in the rDNA operon. Like the PCR-based diagnosis evaluated here, ITS sequencing may provide a generic diagnosis as previously reported by other groups (Estrada-Barcenas *et al.* 2014, Landaburu *et al.* 2014, Irinyi *et al.* 2015) and may supplement traditional morphology-based taxonomy. Indeed, the definitive diagnosis of histoplasmosis can be difficult when studying tissue sections, so the ITS-based identification can play an important role because it will differentiate *Histoplasma* from other morphologically similar agents such as the small variant of *Blastomyces*, or the capsule-deficient cryptococci (Guimarães *et al.* 2006, Guarner & Brandt 2011). The lack of resolution of ITS has also been reported for other taxa and prevents its universal application (Kiss 2012). For some fungi, the second largest subunit of ribosomal polymerase II (*RPB2*) and another recently introduced marker, mini-chromosome maintenance protein (*MCM7*), were utilised with success (Stielow *et al.* 2015).

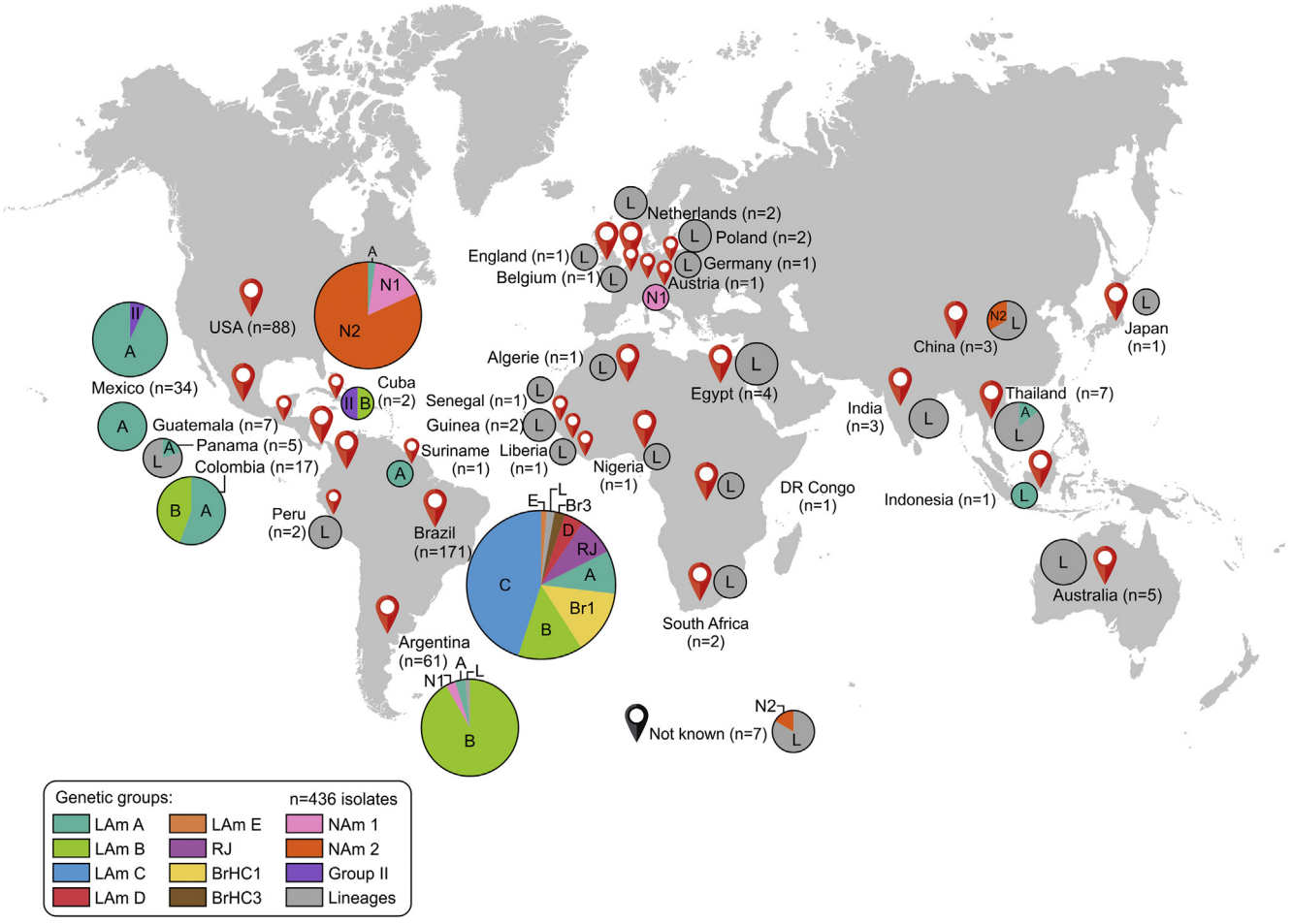


Fig. 10. Global distribution patterns of 436 *Histoplasma* spp. isolates based on DNA sequencing. The sizes of circumferences are roughly proportional to the numbers of strains included. Codes reported within the pies denote genetic groups. Further information about isolate source and GenBank accession number can be found in Table S1.

This reinforces the urgent need to search for alternative reliable barcode markers to improve resolution and unambiguously identify classic and new *Histoplasma* phylogenetic species during routine diagnosis. Therefore, the availability of a "gold standard" phylogeny will make it possible to evaluate the informativeness of other DNA fingerprinting systems for phylogenetic analysis and species recognition. As an alternative, we introduced AFLP genotyping to assess the genetic diversity of Brazilian *Histoplasma* isolates down to the species level. Our results using AFLP markers matched the major genetic group phylogeny assigned by arf, ole and H-anti gene polymorphisms.

AFLP and MLSA indicated negligible gene flow among LAm D, LAm E and RJ, as well between this group ('population 2') and LAm C ('population 1'). Hybrid speciation and introgressive hybridisation occurs more readily between allopatric species and has been suggested to be an important evolutionary force that generates diversity by the recombination of genetic material among divergent lineages. In *Histoplasma*, whole genomic sequencing has been used to demonstrate the occurrence of recent admixture (Sepúlveda *et al.* 2017). Extensive backcrossing between hybrid and the parental species can sometimes lead to introgression if foreign genetic material is integrated into the genomes of either parent (Baack & Rieseberg 2007, Fogelqvist *et al.* 2015). Such important mechanisms whereby new species may emerge can increase fitness and virulence traits and lead to the development of reproductive isolation, a phenomenon often demonstrated among pathogenic fungi (Stukenbrock 2016). However, introgressed alleles are present in low frequencies in *Histoplasma*, supporting the view that they are deleterious (Maxwell *et al.* 2018), which may contribute to reproductive barriers.

Neighbor-net, PHI-test and structure analysis supports a recombination history among *Histoplasma* clades, as well as admixture in population structure with clear evidence of hybrids between clusters. This could have influenced the evolution, radiation, and potentially the emergence of new genotypes of *Histoplasma* in Northeast Brazil. Alternatively, conflicting phylogenetic signals may reflect maintenance of ancestral polymorphism, whereby incomplete lineage sorting leads to ambiguous phylogenetic relationships even when genetic isolation occurs instantly for all genes (Retchless & Lawrence 2010). The use of comparative genomics is necessary to disentangle these competing hypotheses.

In fungi, both sexual (e.g., mating) and asexual origins (e.g., fusion of cells or hyphae) are evident in the emerging list of apparent fungal hybrids (Schardl & Craven 2003, Kohn 2005). Therefore, we assessed the distribution and frequencies of the two mating types in Brazilian populations of *Histoplasma* in an attempt to understand sexual reproduction modes. Since *H. capsulatum* and related species are heterothallic, and our analyses showed equal frequencies of mating types idiomorphs (1:1), we assume that sexual reproduction is a common event among Brazilian *Histoplasma* isolates. In *H. capsulatum s.l.*, some studies have revealed that mating type idiomorphs are equally distributed in the soil but deviate in clinical samples, with the majority being *MAT1-1* (Kwon-Chung *et al.* 1974, Kwon-Chung *et al.* 1984) or *MAT 1-2* (Rodríguez-Arellanes *et al.* 2013), which are usually associated with the geographical origin of the isolates (Muniz *et al.* 2014). Nevertheless, skewed *MAT* loci distribution has been demonstrated for *H. capsulatum* var. *duboisii* in Africa (Valero *et al.* 2018) which may be related to paucity of sexual reproduction and/or strong selection for pleiotropic effects of a mating type allele (Nieuwenhuis & James 2016) or a phenomenon of small populations (Valero *et al.* 2018). Therefore, it is tempting to hypothesise that different *Histoplasma* spp. may have developed a mixed mode of reproduction, ranging from rare to frequent sex, depending on the species and the population characteristics. However, this hypothesis of species-specific reproductive modes in *Histoplasma* should be further investigated in a bigger number of strains.

Notably, the mating type locus has previously been associated with differential virulence in medically relevant fungi (Xu *et al.* 2017). This holds true for sexual pathogens such as *Cryptococcus neoformans* (Kwon-Chung *et al.* 1992), *Aspergillus fumigatus* (Cheema & Christians 2011), and *Mucor irregularis* (Xu *et al.* 2017). However, mating type appears to have no influence on the virulence process of *H. capsulatum s.l.* because no significant difference was found between strains *MAT1-1* and *MAT1-2* in a murine model of infection (Kwon-Chung & Hill 1981). A similar conclusion was drawn for the animal-borne *Sporothrix brasiliensis*, where both *MAT1-1* and *MAT1-2* strains were observed to be highly virulent (Della Terra *et al.* 2017), but deviated in *MAT* frequency in genetic population studies (Teixeira *et al.* 2015).

Using a large number of isolates from geographically and ecologically diverse regions, our study allowed us to address several fundamental questions about the epidemiology of *Histoplasma*. Our findings go beyond those previously reported by others, since we were able to recognise new phylogenetic lineages. Not surprisingly, we identified high genetic diversity and a structured population of *Histoplasma* in Brazil, matching the different geographic sources. Our findings significantly broaden the area of histoplasmosis occurrence, an important feature of emerging pathogens. Therefore, the correct diagnosis will be fundamental to elevate the epidemiological status in one of the most important systemic mycoses of the Americas. Moreover, with the description of new genotypes, we highlight that ITS is not sufficient for routine distinction of classic and new *Histoplasma* species or taxonomic varieties, although it can provide a genus-level identification, which is crucial to guide appropriate clinical therapy. Species-specific reproductive modes in *Histoplasma* ranging from rare to frequent sex may drive dissimilar genetic diversity in this widely distributed pathogen, a hypothesis that should be further investigated in a substantial number of globally sampled strains. From a practical point of view, our data point to the emergence of histoplasmosis caused by a plethora of genotypes and will allow the future proposal of public policies to contain the spread of histoplasmosis.

Limitations of our study include the number of samples associated with clinical status and that MLSA was based only on a few genes. Although those markers have been well characterised by other groups (Kasuga *et al.* 1999, Kasuga *et al.* 2003), we believe that phylogenomics can greatly improve recognition of species, speciation events and evolution of *Histoplasma* (Sepúlveda *et al.* 2017). With the description of such diversity, we open avenues for future comparative genomic studies, which will hopefully lead to a consensus taxonomy, improve understanding of the presence of hybrids in natural populations of medically relevant fungi, test reproductive barriers and explore the significance of these variations.

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APPENDIX A. SUPPLEMENTARY DATA

Supplementary data to this article can be found online at https://doi.org/10.1016/j.simyco.2020.02.001.

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